

Operations on polynomials and series

Problems in competitive programming, especially the ones involving enumeration some kind, are often solved by reducing the problem to computing something on polynomials and formal power series.

This includes concepts such as polynomial multiplication, interpolation, and more complicated ones, such as polynomial logarithms and exponents. In this article, a brief overview of such operations and common approaches to them is presented.

Basic Notion and facts

In this section, we focus more on the definitions and "intuitive" properties of various polynomial operations. The technical details of their implementation and complexities will be covered in later sections.

Polynomial multiplication

Definition

Univariate polynomial is an expression of form $A(x) = a_0 + a_1 x + \dots + a_n x^n$.

The values $a_0, \dots, a_n$ are polynomial coefficients, typically taken from some set of numbers or number-like structures. In this article, we assume that the coefficients are taken from some [field](#), meaning that operations of addition, subtraction, multiplication and division are well-defined for them (except for division by 0) and they generally behave in a similar way to real numbers.

Typical example of such field is the field of remainders modulo prime number p .

For simplicity we will drop the term *univariate*, as this is the only kind of polynomials we consider in this article. We will also write A instead of $A(x)$ wherever possible, which will be understandable from the context. It is assumed that either $a_n \neq 0$ or $A(x)=0$.